Admin

Attributes

- serialVersionUID: long
Operations
+ toString(): String

AdminService

Attributes
- database: Database
- userService: UserService
- authService: AuthService

Operations
+ printAllUsers(): void
+ addUser(User): boolean
- isAdmin(): boolean
+ printAllLogs(): void
- log(String): void
- requireAdmin(): void
+ updateUserEmail(String, String): boolean
+ updateUserFullName(String, String): boolean
+ listAllUsers(): List<User>
+ updateUserLanguage(String, Language): boolean
+ removeUser(String): boolean
+ viewLogsForUser(User): List<LogRecord>
+ viewAllLogs(): List<LogRecord>

AttendanceStatus

Attributes
+ LATE:
+ ABSENT:
+ PRESENT:

Operations
+ valueOf(String): AttendanceStatus
+ values(): AttendanceStatus[]

AuthService

Attributes
- database: Database
- currentUser: User

Operations
+ login(String, String): User
+ isLoggedIn(): boolean
+ logout(): void
+ getCurrentUser(): User
+ hasRole(Class<?>): boolean

Comment

Attributes

- serialVersionUID: long
- text: String
- author: User
- date: Date

Operations
+ hashCode(): int
+ getAuthor(): User
+ getDate(): Date
+ toString(): String
+ getText(): String
+ editText(String): void
+ equals(Object): boolean

Complaint

Attributes

- text: String
- student: Student
- id: int
- idCnt: int
- urgency: UrgencyLevel
- date: Date
- serialVersionUID: long
- teacher: Teacher

Operations
+ setUrgency(UrgencyLevel): void
+ getUrgency(): UrgencyLevel
+ toString(): String
+ hashCode(): int
+ setTeacher(Teacher): void
+ synchronizeNextId(int): void
+ setText(String): void
+ getDate(): Date
+ getId(): int
+ getTeacher(): Teacher
+ getStudent(): Student
+ setStudent(Student): void
+ getText(): String
+ setDate(Date): void
+ equals(Object): boolean

Course

Attributes
- credits: int
- instructorsByLessonType: Map<LessonType, List<Teacher>>
- language: Language
- serialVersionUID: long
- instructors: List<Teacher>
- openForRegistration: boolean
- intendedMajor: String
- idCounter: int
- id: int
- courseCode: String
- type: CourseType
- intendedYear: int
- name: String
- enrolledStudents: List<Student>
- lessons: List<Lesson>

Operations
+ getName(): String
+ getType(): CourseType
- initializeInstructorCollections(): void
+ enrollStudent(Student): void
+ synchronizeIdCounter(int): void
+ getInstructors(): List<Teacher>
+ addInstructor(Teacher): void
+ getIntendedMajor(): String
+ addInstructor(Teacher, LessonType): void
+ getCredits(): int
+ getIntendedYear(): int
+ equals(Object): boolean
+ setIntendedMajor(String): void
- addInstructorToAllInstructors(Teacher): void
+ getAllInstructors(): List<Teacher>
+ getCourseCode(): String
+ hashCode(): int
+ getLessons(): List<Lesson>
+ getLanguage(): Language
+ getId(): int
+ getEnrolledStudents(): List<Student>
+ isOpenForRegistration(): boolean
+ setOpenForRegistration(boolean): void
+ getInstructorsByLessonType(LessonType): List<Teacher>
+ addLesson(Lesson): void
+ toString(): String
+ setIntendedYear(int): void
+ removeStudent(Student): void

CourseRegistration

Attributes

- serialVersionUID: long
- student: Student
- course: Course
- idCounter: int
- reviewedBy: Manager
- rejectionReason: String
- createdDate: Date
- status: RegistrationStatus
- id: int
- reviewedDate: Date

Operations
+ getCourse(): Course
+ getId(): int
+ getStatus(): RegistrationStatus
+ reject(Manager, String): void
+ synchronizeIdCounter(int): void
+ equals(Object): boolean
+ getRejectionReason(): String
+ getReviewedDate(): Date
+ getCreatedDate(): Date
+ ensureId(): void
+ getReviewedBy(): Manager
+ getStudent(): Student
+ hashCode(): int
+ approve(Manager): void
+ toString(): String

CourseType

Attributes

+ FREE_ELECTIVE:
+ MAJOR:
+ MINOR:

Operations
+ values(): CourseType[]
+ valueOf(String): CourseType

Database

Attributes
- users: List<User>
- journals: List<Journal>
- techSupportReqs: List<TechSupportReq>
- lessons: List<Lesson>
- researchProjects: List<ResearchProject>
- logs: List<LogRecord>
- instance: Database
- news: List<News>
- complaints: List<Complaint>
- courseRegistrations: List<CourseRegistration>
- studentOrganizations: List<StudentOrganization>
- reports: List<Report>
- nextTechSupportReqId: int
- messages: List<Message>
- courses: List<Course>
- researchPapers: List<ResearchPaper>
- serialVersionUID: long
Operations
+ getReports(): List<Report>
+ addCourse(Course): void
+ findNewsByTopic(NewsTopic): List<News>
+ removeUser(User): boolean
+ addResearchProject(ResearchProject): void
+ findResearchProjectByTopic(String): ResearchProject
+ findUserById(int): User
+ findComplaintsByStudent(Student): List<Complaint>
+ save(): void
+ getLessons(): List<Lesson>
+ findTechSupportReqById(int): TechSupportReq
+ findResearchPaperByDoi(String): ResearchPaper
+ getTechSupportReqs(): List<TechSupportReq>
+ findComplaintsByTeacher(Teacher): List<Complaint>
+ addMessage(Message): void
+ findRegistrationsByStatus(RegistrationStatus): List<CourseRegistration>
+ getInstance(): Database
+ findUserByUsername(String): User
+ findMessageById(int): Message
+ findMessagesBySender(Employee): List<Message>
+ getMessages(): List<Message>
+ findCourseByName(String): Course
+ getJournals(): List<Journal>
+ findCourseRegistrationById(int): CourseRegistration
+ removeNews(News): boolean
+ getResearchPapers(): List<ResearchPaper>
+ getCourses(): List<Course>
+ findLogById(int): LogRecord
+ addStudentOrganization(StudentOrganization): void
+ addNews(News): void
+ removeReport(Report): boolean
+ setInstance(Database): void
+ findTechSupportReqsByStatus(RequestStatus): List<TechSupportReq>
+ getStudentOrganizations(): List<StudentOrganization>
+ findResearchPaperByTitle(String): ResearchPaper
+ findReportByTitle(String): Report
+ addReport(Report): void
+ findCourseByCode(String): Course
+ getResearchProjects(): List<ResearchProject>
+ addLog(LogRecord): void
+ addCourseRegistration(CourseRegistration): void
+ addResearchPaper(ResearchPaper): void
- ensureLists(): void
+ findStudentOrganizationByName(String): StudentOrganization
+ findLessonsByCourse(Course): List<Lesson>
+ addLesson(Lesson): void
+ addTechSupportReq(TechSupportReq): void
+ findJournalByName(String): Journal
+ findLogsByUser(User): List<LogRecord>
+ nextTechSupportReqId(): int
+ load(): void
+ getLogs(): List<LogRecord>
+ findRegistrationsByStudent(Student): List<CourseRegistration>
+ getUsers(): List<User>
+ getComplaints(): List<Complaint>
+ findComplaintById(int): Complaint
+ addJournal(Journal): void
- normalizeAfterLoad(): void
- synchronizeGeneratedIds(): void
+ getCourseRegistrations(): List<CourseRegistration>
+ addComplaint(Complaint): void
+ findMessagesByReceiver(Employee): List<Message>
+ getNews(): List<News>
+ removeMessage(Message): boolean
+ addUser(User): void
+ findNewsByTitle(String): News
+ findLessonById(int): Lesson

DegreeType

Attributes
+ PHD:
+ MASTER:

Operations
+ values(): DegreeType[]
+ valueOf(String): DegreeType

Employee

Attributes

- employeeId: String
- salary: double
- serialVersionUID: long
- department: String
- hierDate: Date

Operations
+ getDepartment(): String
+ getSalary(): double
+ getEmployeeId(): String
+ toString(): String
+ getHierDate(): Date
+ getRoleName(): String

FileStorage

Attributes

- FILE_NAME: String
Operations
+ load(): Database
- getDatabaseFile(): File
+ save(Database): void

Format

Attributes
+ BIBTEX:
+ PLAIN_TEXT:
Operations
+ values(): Format[]
+ valueOf(String): Format

GraduateStudent

Attributes

- papers: List<ResearchPaper>
- degreeType: DegreeType
- supervisor: Researcher
- serialVersionUID: long
- diplomaProjects: List<ResearchPaper>
- projects: List<ResearchProject>
Operations
+ setDegreeType(DegreeType): void
+ calculateHIndex(): int
+ getDegreeType(): DegreeType
+ getProjects(): List<ResearchProject>
+ addPaper(ResearchPaper): void
+ getPapers(): List<ResearchPaper>
+ getDiplomaProjects(): List<ResearchPaper>
+ setSupervisor(Researcher): void
+ addDiplomaProject(ResearchPaper): void
+ toString(): String
+ getSupervisor(): Researcher
+ addProject(ResearchProject): void

I18n

Attributes
- bundle: ResourceBundle
- currentLanguage: Language
Operations
+ setLanguage(Language): void
+ t(String, Object[]): String
+ getCurrentLanguage(): Language

InvalidSupervisorEx

Attributes

Operations

Journal

Attributes
- idCounter: int
- papers: List<ResearchPaper>
- name: String
- serialVersionUID: long
- id: int
- subscribers: List<User>

Operations
+ getId(): int
+ synchronizeIdCounter(int): void
+ unsubscribe(User): void
+ getPapers(): List<ResearchPaper>
+ getName(): String
+ getSubscribers(): List<User>
+ equals(Object): boolean
+ subscribe(User): void
+ addPaper(ResearchPaper): void
+ hashCode(): int
+ toString(): String
+ ensureId(): void

JournalService

Attributes
- database: Database
- authService: AuthService

Operations
+ getAllJournals(): List<Journal>
+ notifySubscribers(Journal): void
+ findJournalByName(String): Journal
+ publishPaper(Journal, ResearchPaper): void
+ addJournal(Journal): void
+ subscribe(User, Journal): void
+ printJournalInfo(Journal): void
- log(String): void
+ unsubscribe(User, Journal): void

Language

Attributes
+ RU:
+ KZ:
+ EN:

Operations
+ valueOf(String): Language
+ values(): Language[]

Lesson

Attributes
- idCounter: int
- dayOfWeek: String
- room: String
- course: Course
- teacher: Teacher
- id: int
- serialVersionUID: long
- timeSlot: String
- type: LessonType
Operations
+ getRoom(): String
+ getType(): LessonType
+ getTeacher(): Teacher
+ getCourse(): Course
+ getId(): int
+ getTimeSlot(): String
+ getDayOfWeek(): String
+ synchronizeIdCounter(int): void
+ equals(Object): boolean
+ hashCode(): int
+ toString(): String

LessonType

Attributes
+ PRACTICE:
+ LECTURE:

Operations
+ values(): LessonType[]
+ valueOf(String): LessonType

LogRecord

Attributes
- id: int
- user: User
- idCnt: int
- action: String
- date: Date
- serialVersionUID: long
Operations
+ getUser(): User
+ getId(): int
+ getAction(): String
+ getDate(): Date
+ toString(): String
+ synchronizeNextId(int): void

Manager

Attributes

- managerType: ManagerType
- serialVersionUID: long
Operations
+ setManagerType(ManagerType): void
+ getManagerType(): ManagerType
+ toString(): String

ManagerService

Attributes

- MAX_STUDENT_CREDITS: int
- authService: AuthService
- database: Database
- reportService: ReportService
Operations
+ assignCourseToTeacher(String, int, LessonType): void
+ rejectRegistration(int, String): boolean
+ getPendingRegistrations(): List<CourseRegistration>
- hasLessonConflict(Course, Lesson): boolean
+ viewTeachersAlphabetically(List<Teacher>): List<Teacher>
+ removeNews(String): boolean
- requireManager(): Manager
+ addNews(News): void
+ addLessonToCourse(Course, Lesson): boolean
- registerStudentToCourse(Student, Course): void
+ viewStudentsSortedByGpa(List<Student>): List<Student>
- log(String): void
+ createAcademicReport(List<Student>): Report
+ unregisterStudentFromCourse(Student, Course): boolean
+ addCourseForRegistration(Course): void
+ approveRegistration(int): boolean
+ pinNews(String): void

ManagerType

Attributes
+ DEPARTMENT:
+ OR:

Operations
+ values(): ManagerType[]
+ valueOf(String): ManagerType

Mark

Attributes
- firstAttestation: double
- finalExam: double
- student: Student
- course: Course
- totalScore: double
- serialVersionUID: long
- idCounter: int
- secondAttestation: double
- id: int

Operations
+ toString(): String
+ setFirstAttestation(double): void
+ getFinalExam(): double
- validateScore(double): double
+ setFinalExam(double): void
+ equals(Object): boolean
+ getTotalScore(): double
+ getStudent(): Student
+ getCourse(): Course
+ setSecondAttestation(double): void
+ getSecondAttestation(): double
- calculateTotal(): void
+ hashCode(): int
+ synchronizeIdCounter(int): void
+ getFirstAttestation(): double
+ getId(): int

MarkException

Attributes

Operations

MenuPrinter

Attributes
- WIDTH: int

Operations
+ printPromptBox(String): void
- trim(String): String
- printSeparator(): void
+ print(String, String, List<String>): void
- printBottom(): void
- printTop(): void
- printCentered(String): void
- printLine(String): void

Message

Attributes
- idCnt: int
- sentDate: Date

- sender: Employee
- isRead: boolean
- serialVersionUID: long
- text: String
- receiver: Employee
- id: int

Operations
+ equals(Object): boolean
+ markAsRead(): void
+ getText(): String
+ getSentDate(): Date
+ hashCode(): int
+ getReceiver(): Employee
+ getIdCnt(): int
+ toString(): String
+ getSender(): Employee
+ isRead(): boolean
+ getId(): int
+ synchronizeNextId(int): void

MessageService

Attributes
- authService: AuthService
- database: Database

Operations
+ getSentMessages(): List<Message>
+ searchMessages(String): List<Message>
+ printSentMessages(): void
- requireEmployee(): Employee
+ openMessage(int): Message
+ sendMessage(Employee, String): Message
+ getInboxFrom(Employee): List<Message>
+ getInbox(): List<Message>
- log(String): void
+ markAllRead(): void
+ getUnreadMessages(): List<Message>
+ deleteMessage(int): boolean
+ getPreview(Message): String
+ printInbox(): void
+ getUnreadMessagesCount(): int

News

Attributes

- pinned: boolean
- content: String
- topic: NewsTopic
- idCounter: int
- date: Date
- id: int
- serialVersionUID: long
- title: String
- comments: List<Comment>

Operations
+ getComments(): List<Comment>
+ getDate(): Date
+ getId(): int
+ hashCode(): int
+ synchronizeIdCounter(int): void
+ pin(): void
+ ensureId(): void
+ addComment(Comment): void
+ equals(Object): boolean
+ toString(): String
+ getTitle(): String
+ unpin(): void
+ getTopic(): NewsTopic
+ getContent(): String
+ isPinned(): boolean

NewsService

Attributes
- instance: NewsService
- database: Database
- authService: AuthService

Operations
+ pinNews(News): void
+ getAllNews(): List<News>
+ findNewsByTitle(String): News
- requireLoggedIn(): User
- db(): Database
+ getNewsByTopic(NewsTopic): List<News>
- requireNewsManager(): void
- log(String): void
+ addNews(News): void
+ addComment(News, Comment): void
- requireResearchNewsAuthor(): void
+ getInstance(): NewsService
+ createResearchNews(String, String): void

NewsTopic

Attributes
+ ACADEMIC:
+ RESEARCH:
+ ANNOUNCEMENT:
+ EVENT:

Operations
+ values(): NewsTopic[]
+ valueOf(String): NewsTopic

NotResearcherEx

Attributes

Operations

RegistrationStatus

Attributes
+ APPROVED:
+ REJECTED:
+ PENDING:

Operations
+ values(): RegistrationStatus[]
+ valueOf(String): RegistrationStatus

Report

Attributes

- reportContent: String
- id: int
- title: String
- studentData: List<Student>
- serialVersionUID: long
- idCounter: int

Operations
+ getId(): int
+ getReportContent(): String
+ synchronizeIdCounter(int): void
+ ensureId(): void
+ getTitle(): String
+ toString(): String
+ getStudentData(): List<Student>
+ hashCode(): int
+ equals(Object): boolean

ReportService

Attributes
- database: Database
- authService: AuthService

Operations
+ printReport(Report, Format): void
- calculateGpaFromTranscript(Transcript): double
- printBibtex(Report): void
+ getReportByTitle(String): Report
- printPlainText(Report): void
+ createAcademicReport(List<Student>): Report
- requireManager(): void
+ deleteReport(String): boolean
+ createResearchReport(List<Teacher>): Report
+ getAllReports(): List<Report>
- log(String): void

RequestStatus

Attributes
+ ACCEPTED:
+ NEW:
+ VIEWED:
+ DONE:
+ REJECTED:

Operations
+ valueOf(String): RequestStatus
+ values(): RequestStatus[]

ResearchMetrics

Attributes

Operations
+ calculateHIndex(List<ResearchPaper>): int

ResearchPaper

Attributes

- journal: Journal
- doi: String
- publishDate: Date
- serialVersionUID: long
- authors: List<Researcher>
- pages: int
- citations: int
- idCounter: int
- id: int
- title: String
Operations
+ getPublishDate(): Date
+ getTitle(): String
+ toString(): String
+ getPages(): int
+ getCitations(): int
+ synchronizeIdCounter(int): void
+ getDoi(): String
+ getAuthors(): List<Researcher>
- validateNonNegative(int, String): int
+ equals(Object): boolean
+ hashCode(): int
+ addCitation(): void
+ getId(): int
+ ensureId(): void
+ getJournal(): Journal

ResearchPaperCitationComparator

Attributes

Operations
+ compare(ResearchPaper, ResearchPaper): int

ResearchPaperDateComparator

Attributes

Operations
+ compare(ResearchPaper, ResearchPaper): int

ResearchPaperLengthComparator

Attributes

Operations
+ compare(ResearchPaper, ResearchPaper): int

ResearchPaperService

Attributes
- authService: AuthService
- journalService: JournalService
- database: Database

Operations
+ getAllJournals(): List<Journal>
- log(String): void
+ printPapers(Researcher, Comparator<ResearchPaper>): void
+ addPaperToDatabase(ResearchPaper): void
+ getCitation(ResearchPaper, Format): String
+ getAllPapers(): List<ResearchPaper>
+ addDiplomaPaper(GraduateStudent, ResearchPaper): boolean
+ printPaperCitation(ResearchPaper, Format): void
- getPapersByResearcher(Researcher): List<ResearchPaper>
+ publishPaper(Researcher, ResearchPaper, Journal): void
+ printPaperInfo(ResearchPaper): void
+ findJournalByName(String): Journal
- requireResearcher(): User

ResearchProject

Attributes

- participants: List<Researcher>
- serialVersionUID: long
- idCounter: int
- publishedPapers: List<ResearchPaper>
- topic: String
- id: int

Operations
+ removeParticipant(Researcher): void
+ synchronizeIdCounter(int): void
+ addPublishedPaper(ResearchPaper): void
+ getTopic(): String
+ hashCode(): int
+ getPublishedPapers(): List<ResearchPaper>
+ addParticipant(Researcher): void
+ toString(): String
+ getParticipants(): List<Researcher>
+ getId(): int
+ ensureId(): void
+ equals(Object): boolean

ResearchProjectService

Attributes
- database: Database
- authService: AuthService

Operations
+ findProjectByTopic(String): ResearchProject
+ printProjectInfo(ResearchProject): void
- requireManager(): Manager
+ getAllProjects(): List<ResearchProject>
+ joinProject(ResearchProject, Researcher): void
- requireResearcher(): User
+ addProject(ResearchProject): void
+ assignSupervisor(GraduateStudent, Researcher): void
+ addPaperToProject(ResearchProject, ResearchPaper): void
- log(String): void

ResearchService

Attributes
- database: Database
- instance: ResearchService
- authService: AuthService

Operations
+ findPaperByDoi(String): ResearchPaper
- db(): Database
+ addPaper(ResearchPaper): void
+ getInstance(): ResearchService
- requireLoggedIn(): User
- log(String): void
+ findPaperByTitle(String): ResearchPaper
+ getAllPapers(): List<ResearchPaper>
- requireResearchMutationAccess(): void

Researcher

<<interface>>

Attributes

Operations
+ calculateHIndex(): int
+ addPaper(ResearchPaper): void
+ addProject(ResearchProject): void
+ printPapers(Comparator<ResearchPaper>): void
+ getProjects(): List<ResearchProject>
+ getPapers(): List<ResearchPaper>

Student

Attributes

- minor: String
- school: String
- failedCoursesCount: int
- registeredCourses: List<Course>
- transcript: Transcript
- organizations: List<StudentOrganization>
- studentId: String
- yearOfStudy: int
- gpa: double
- serialVersionUID: long
- major: String
- credits: int

Operations
+ getGpa(): double
+ setCredits(int): void
+ toString(): String
+ setGpa(double): void
+ removeRegisteredCourse(Course): void
+ setSchool(String): void
+ hashCode(): int
+ leaveOrganization(StudentOrganization): void
+ getYearOfStudy(): int
+ setMajor(String): void
+ setMinor(String): void
+ setYearOfStudy(int): void
+ getMinor(): String
+ getOrganizations(): List<StudentOrganization>
+ setFailedCoursesCount(int): void
+ getSchool(): String
+ getStudentId(): String
+ getMajor(): String
+ getFailedCoursesCount(): int
+ setStudentId(String): void
+ getCredits(): int
+ getTranscript(): Transcript
+ joinOrganization(StudentOrganization): void
+ addRegisteredCourse(Course): void
+ equals(Object): boolean
+ getRegisteredCourses(): List<Course>

StudentGpaComparator

Attributes

Operations
+ compare(Student, Student): int

StudentMenu

Attributes
userService: UserService
sc: Scanner
studentService: StudentService
authService: AuthService
journalService: JournalService
transcriptService: TranscriptService
Operations
unsubscribeJournal(Student): void
printMarks(List<Mark>): void
readLanguage(): Language
registerForCourse(): void
printOrganizations(List<StudentOrganization>): void
rateTeacher(): void
printCourses(List<Course>, String): void
readInt(String): int
printJournals(): void
readJournal(): Journal
readDouble(String): double
viewTeacherInfo(): void
printNotifications(Student): void
+ show(): void
# subscribeJournal(Student): void
# promptRequired(String): String
# journalsMenu(Student): void
# switchLanguage(Student): void
# organizationsMenu(Student): void

StudentOrganization

Attributes
- id: int
- head: Student
- idCounter: int
- members: List<Student>
- name: String
- serialVersionUID: long
- description: String
Operations
+ addMember(Student): void
+ synchronizeIdCounter(int): void
+ getId(): int
+ setHead(Student): void
+ isMember(Student): boolean
+ getName(): String
+ getDescription(): String
+ getHead(): Student
+ removeMember(Student): void
+ getMembers(): List<Student>
+ toString(): String
+ equals(Object): boolean
+ hashCode(): int

StudentService

Attributes
- database: Database
- MAX_STUDENT_CREDITS: int
- authService: AuthService

Operations
+ registerForCourse(String): boolean
+ getAllOrganizations(): List<StudentOrganization>
- hasPendingRegistration(Student, Course): boolean
- log(String): void
+ joinOrganization(String): boolean
- getPendingCredits(Student): int
+ getAvailableCourses(): List<Course>
- requireStudent(): Student
+ leaveOrganization(String): boolean
+ getTeacherInfo(String): Teacher
+ rateTeacher(int, double): boolean
- isCourseAvailableForStudent(Course, Student): boolean

Teacher

Attributes

- papers: List<ResearchPaper>
- teacherType: TeacherType
- projects: List<ResearchProject>
- rating: double
- complaints: List<Complaint>
- serialVersionUID: long
- ratingCount: int
- courses: List<Course>

Operations
+ toString(): String
+ getComplaints(): List<Complaint>
+ addPaper(ResearchPaper): void
+ getPapers(): List<ResearchPaper>
+ getCourses(): List<Course>
+ addCourse(Course): void
+ calculateHIndex(): int
+ getTeacherType(): TeacherType
+ addProject(ResearchProject): void
+ getProjects(): List<ResearchProject>
+ addRating(double): void
+ addComplaint(Complaint): void
+ getRating(): double

TeacherNameComparator

Attributes

Operations
+ compare(Teacher, Teacher): int

TeacherService

Attributes
- authService: AuthService
- FAILING_SCORE: double
- database: Database

Operations
+ putMark(Student, Course, Mark): boolean
+ assignCourse(Course, LessonType): void
+ viewStudents(Course): List<Student>
+ assignCourse(Course): void
+ sendComplaint(Student, UrgencyLevel, String): Complaint
- log(String): void
- requireTeacher(): Teacher

TeacherType

Attributes
+ PROFESSOR:
+ TUTOR:
+ LECTOR:
+ SENIOR_LECTOR:
Operations
+ valueOf(String): TeacherType
+ values(): TeacherType[]

TechSupportReq

Attributes

- sender: Employee
- assignedSpecialist: TechSupportSpecialist
- createdDate: Date
- id: int
- serialVersionUID: long
- description: String
- rejectionReason: String
- resolvedDate: Date
- status: RequestStatus
Operations
+ assignTo(TechSupportSpecialist): void
+ getId(): int
+ getSender(): Employee
+ getStatus(): RequestStatus
+ setStatus(RequestStatus): void
+ getRejectionReason(): String
+ getDescription(): String
+ toString(): String
+ getResolvedDate(): Date
+ hashCode(): int
+ getAssignedSpecialist(): TechSupportSpecialist
+ equals(Object): boolean
+ markDone(): void
+ getCreatedDate(): Date
+ reject(String): void

TechSupportService

Attributes
- authService: AuthService
- database: Database

Operations
+ getRequestsByStatus(RequestStatus): List<TechSupportReq>
+ printMyRequests(): void
- requireSupportViewer(): void
+ acceptRequest(int): boolean
- log(String): void
+ submitRequest(String): TechSupportReq
- requireEmployee(): Employee
+ getMyRequests(): List<TechSupportReq>
+ viewNewRequests(): List<TechSupportReq>
- findById(int): TechSupportReq
- requireSpecialist(): TechSupportSpecialist
+ markAsDone(int): boolean
+ viewRequest(int): TechSupportReq
+ printAllRequests(): void
+ rejectRequest(int, String): boolean

TechSupportSpecialist

Attributes

- serialVersionUID: long
Operations
+ toString(): String

Transcript

Attributes
- student: Student
- marks: List<Mark>
- failedAttemptsByCourse: Map<Course, Integer>
- serialVersionUID: long
- MAX_FAILED_ATTEMPTS_PER_COURSE: int
Operations
+ getStudent(): Student
+ setStudent(Student): void
+ calculateGpa(): double
+ setMarks(List<Mark>): void
+ addMark(Mark): void
+ equals(Object): boolean
- ensureFailedAttemptsByCourse(): Map<Course, Integer>
+ incrementFailedAttempts(Course): void
+ hashCode(): int
+ getMarks(): List<Mark>
+ getFailedAttempts(Course): int
+ hasExceededFailedAttempts(Course): boolean
+ toString(): String

TranscriptService

Attributes
- database: Database
- authService: AuthService

Operations
+ printTranscript(Transcript): void
- requireTranscriptAccess(Transcript): void

UrgencyLevel

Attributes
+ HIGH:
+ MEDIUM:
+ LOW:

Operations
+ values(): UrgencyLevel[]
+ valueOf(String): UrgencyLevel

User

Attributes

- email: String
- id: int
- username: String
- idCounter: int
- language: Language
- notifications: List<String>
- serialVersionUID: long
- password: String
- fullName: String
Operations
+ setLanguage(Language): void
+ getNotifications(): List<String>
+ equals(Object): boolean
+ synchronizeIdCounter(int): void
+ getUsername(): String
+ hashCode(): int
+ setEmail(String): void
+ getId(): int
+ setFullName(String): void
+ setUsername(String): void
- hashPassword(String): String
+ getLanguage(): Language
+ addNotification(String): void
+ login(String): boolean
+ getEmail(): String
+ toString(): String
+ getFullName(): String

UserNamePadding

Attributes

Operations
+ padRight(String, int): String

UserService

Attributes
- authService: AuthService
- database: Database

Operations
+ getAllStudents(): List<Student>
+ getAllEmployees(): List<Employee>
+ getTeacherInfoForStudent(Student, Course): Teacher
+ registerUser(User): boolean
+ saveCurrentUserNotification(String): void
+ changeLanguage(User, Language): void
- requireAdminOrSelf(User): void
+ changeUsername(User, String): boolean
- requireAdmin(): void
+ getAllUsers(): List<User>
+ getAllTeachers(): List<Teacher>
- log(String): void
+ findById(int): User
+ getAllCourses(): List<Course>
+ removeUser(String): boolean
+ findByUsername(String): User

1

adminService

1

authService padding

userService

1 1

authService

Â«createÂ»

database userService

1 1

database

Â«createÂ»

currentUser

1 1

author

1 1

student teacher

1 1

urgency

1 1

type language

*

lessons

*

instructorsByLessonType

1 *

enrolledStudents

1 *

instructors

1 1

course reviewedBy status

1 1

student

1 *
complaints courses courseRegistrations lessons journals logs messages news reports researchPapers researchProjects studentOrganizations techSupportReqs users

1 1

degreeType

Â«createÂ»
*

diplomaProjects projects

1

supervisor

1 1

currentLanguage

1 *

papers

1 *

subscribers

1 1

authService

1

database

Â«createÂ»

1 1

course

1 1

type
1 1

teacher

1 1

user

1 1

managerType

1

managerService

1

messageService

1

projectService

1

techSupportService

1 1

authService database

Â«createÂ»

reportService

1 1

course

1 1

student

1 1

sender

1 1

authService database

Â«createÂ»

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comments

1

topic

1 1
authService database

Â«createÂ»

1 *

studentData

1 1

authService database

Â«createÂ»

1 1

journal
1 *

authors

1 1

authService database journalService
Â«createÂ»

1 *

publishedPapers

*

participants

1 1

authService database

Â«createÂ»

1 1

authService database

Â«createÂ»

* 1

organizations registeredCourses

Â«createÂ»

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transcript

1 1
authService journalService studentService transcriptService userService

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members

1 1

authService

Â«createÂ»

database

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complaints

1 *

courses

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papers

* 1

projects teacherType

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authService

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database

11

status sender assignedSpecialist

1 1

authService database

Â«createÂ»

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sageService

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hSupportService

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failedAttemptsByCourse

*

marks

1 1

student

1 1 1

authService database

1 1

language

1 1

authService

1

database

Â«createÂ»